

Joshua Brenneman

Myrtle Beach, SC | jbrenn.dev@gmail.com | github.com/jabrennem | linkedin.com/in/joshuabrennemana

Senior Software Engineer focused on AWS-native platforms for biomedical research, clinical genomics, scientific computing, data systems, and machine learning. Experienced in reusable infrastructure, distributed workflows, developer enablement, and reliable cloud operations.

EXPERIENCE

Senior Software Engineer

Apr 2023 - Present

Nationwide Children's Hospital | Office of Data Sciences, Cloud Solutions Team

- Design reusable AWS platform capabilities across more than 10 AWS accounts and an engineering ecosystem of more than 600 repositories.
- Lead architecture and implementation of a research lakehouse spanning storage, ETL, governance, lineage, and visualization with Amazon S3, Athena, and Apache Iceberg.
- Support production workflows for more than five research and clinical laboratories working with petabyte-scale datasets.
- Build orchestration and compute patterns with Step Functions, Lambda, EMR, Glue, Batch, ECS, Docker, and ECR.
- Partner with data scientists on SageMaker workloads and AWS infrastructure for generative-AI applications using Amazon Bedrock.

Bioinformatics Software Developer

Oct 2019 - Apr 2023

Nationwide Children's Hospital | Institute for Genomic Medicine, Cloud Solutions Team

- Designed cloud-native genomics workflows using Step Functions, Lambda, AWS Batch, and containerized next-generation sequencing pipelines.
- Built cloud services for genomic variant analysis and research-data applications with Python and CloudFormation.
- Worked with bioinformatics scientists to translate research workflows into reliable production systems.

Bioinformatics Software Developer

Oct 2017 - Oct 2019

Nationwide Children's Hospital | Institute for Genomic Medicine, Clinical Informatics Team

- Developed applications and backend services supporting clinical-genomics operations and laboratory workflows.
- Translated stakeholder requirements into production software and supported applications through iterative improvement.

Big Data Application Developer

Jun 2016 - Oct 2017

CGI | Consulting engagement with PNC Bank

- Developed Hadoop and Apache Spark data-processing workflows and improved distributed batch-system performance and reliability.

TECHNICAL CAPABILITIES

- **Platform engineering:** AWS platform architecture, serverless systems, distributed workflows, event-driven architecture, developer enablement.
- **Cloud and data:** Amazon S3, Athena, Apache Iceberg, AWS Glue, EMR, ETL, lakehouse architecture.
- **Compute and containers:** AWS Lambda, Batch, ECS, Docker, Amazon ECR, production orchestration patterns.
- **Infrastructure and delivery:** AWS CDK, CloudFormation, AWS SAM, GitHub Actions, CodePipeline, CodeBuild, CI/CD.
- **Software and AI:** Python, TypeScript, JavaScript, modern C++, Amazon SageMaker, Amazon Bedrock, MCP.

EDUCATION AND CERTIFICATION

EDUCATION

B.A. Linguistics
Minor in Computer Science

CERTIFICATION

AWS Certified Developer - Associate